Module ddi: Database-Design and -Implementation

Mini-Challenge for LE2 and LE3: SQL and NoSQL

German: Mini-Challenge zu LE2 und LE3: SQL und NoSQL

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A graded submission is part of the assessment for LE2 and LE3. This mini-challenge can be submitted for this purpose. This mini-challenge may be handed in as a group of a maximum of 2 people. The grade achieved counts as a submission for both, LE2 and LE3.

Task: Comparison of different database types
Objective: A solution with NoSQL and with SQL is compared on the basis of a specific problem.

Specify a problem that can reasonably implemented with a NoSQL database. The problem must be non-trivial, i.e. it should contain at least one foreign key in SQL. Be it between two tables or as a self-reference.
Select a NoSQL database that fits the problem. Determine the advantages of the chosen database and how you can measure these advantages.
Design the conceptual data model and then implement it with the selected NoSQL database and with a relational database. The implementation should be done in such a way that it is optimised for the problem. This means that the logical and physical data model is adapted to the query requirements.
Fill both databases with the same data.
Compare the two databases with the defined measurements.

Report
Write a report in which you reflect the results and answer the following questions:

General questions
What is your prior knowledge of databases from your previous practical experience?
What was the most important insight for you?
Are there any other important points that had an important influence on your learning success?

Specific questions
Which NoSQL database did you choose and why?
What does the problem look like?
What does the conceptual data model look like?
What does the implementation in the two databases look like?
What advantages did you expect from the NoSQL database?
What measurement criteria did you define?
What do the measurement results look like?
Interpret and reflect on these measurement results.

What we expect
The report as a document (Word or PDF-format), containing the conceptual model (hand-draft or created with a tool). The scripts for creating the databases and loading the data (SQL and NoSQL) as separate text file(s).

Criteria for grading
In the report (containing up to 20 pages, text and figures) all questions are answered in a coherent, well-founded and understandable way.